EthioEMR Platform Guide

How the platform is put together, run, tested and changed

v0.1, draft for the platform team Repository `ethioemr-platform`, branch `feat/initial-platform`, 22 October 2026

> STATUS OF THIS DOCUMENT
>
> **v0.1, a draft for engineers and maintainers.** Every number in it was read from the repository or measured on the development laptop on 22 October 2026. Nothing here is reviewed or merged yet. Items that belong to other owners (the distro team, the content team, the AI service owner) are marked as theirs and left open. It is written for engineers who start, change or review the platform; clinicians and form builders use the *EthioEMR User Manual*.

1. What the platform is
2. Architecture
3. Repository layout and ownership
4. Running it
5. Ports
6. Configuration (`.env`)
7. Extra frontend modules (the overlay)
8. AI assistant configuration in the gateway
9. Speech service
10. Form Studio build
11. Testing
12. Known gaps
13. Security notes
14. Conventions
15. Where to see the work

1. What the platform is

The platform is the team's EthioEMR (OpenMRS 3, TenaEHR) exactly as it is on GitHub `main`, plus the modules built on top of it that are not in the published images yet. `distro/` is a byte-for-byte copy of `palladium-group/ethiopiaemr-distro-referenceapplication` at commit `9147129`. Everything specific to the platform lives in one overlay compose file, the gateway template and `modules/`.

Part	What it does	Source
Form Studio	Build, check and export forms. Never writes to a live EMR.	`modules/esm-form-studio-app`
AI assistant	Help & Search, Form Filler, Diagnosis Assistant and the Speak input, inside the patient chart.	`modules/esm-ai-assistant-app/dist` (built bundle only)
Speech service	Local Amharic and English speech-to-text for Speak. Audio stays on the machine.	`services/asr-service`
Demo-video tooling	Scripted, narrated product videos.	`tools/demo-video`

Item	Value
App	`http://localhost:18200/openmrs/spa` (user `admin`, password `Admin123`, location Main Pharmacy)
MySQL	`localhost:3320` (user `openmrs`, password `openmrs`)
App shell	10.0.1-pre.5263 (pinned by the team), Form Builder 3.5.1
Frontend apps	56 in the live `importmap.json` (22 Oct): 54 team apps and 2 overlay modules.
Images	`meronmen/ethiopiaemr-frontend:sha-9147129`, `meronmen/ethiopiaemr-backend:sha-9147129`, `mysql:8.0.45`, gateway built from `distro/gateway`

Synthetic test data only. Never enter real patient information on this platform.

2. Architecture

One gateway on port 18200 decides where each browser request goes. The team's frontend and backend images are used unchanged; the platform adds a module overlay, an AI configuration file and a local speech service.

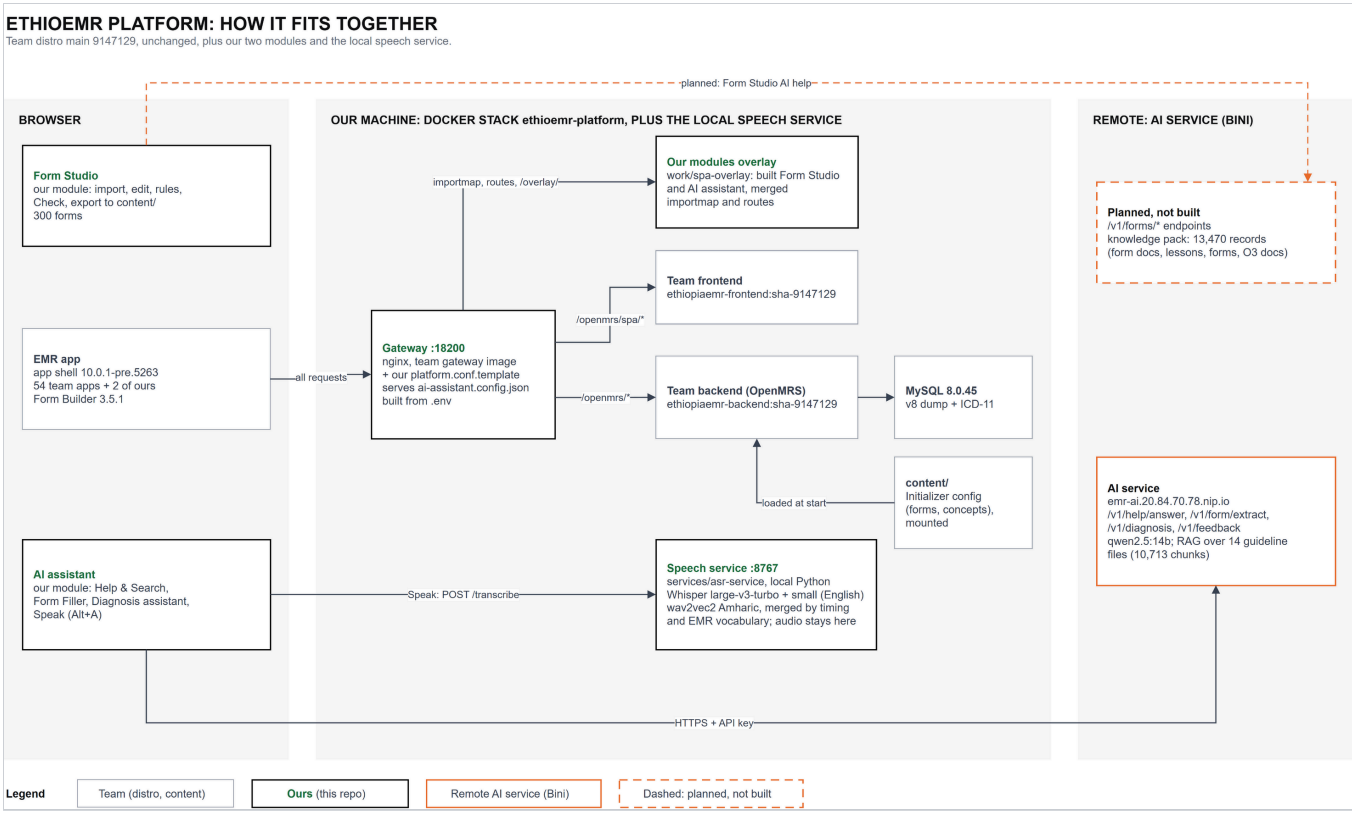


Figure 1. How the platform fits together. Source: `docs/platform/architecture.drawio`. Black boxes are this repository, grey boxes the team's distro and content, orange the remote AI service, dashed boxes planned work.

Request routing in the gateway

The gateway is the team's nginx image with `gateway/platform.conf.template` mounted in place of its default template.

Path	Served from
`/openmrs/spa/importmap.json`, `/openmrs/spa/routes.registry.json`	`work/spa-overlay/`: the team's app list and routes merged with this repo's modules. Never cached.
`/openmrs/spa/overlay/<module>/`	`work/spa-overlay/modules/` : the built Form Studio and AI assistant files.
`/openmrs/spa/ai-assistant.config.json`	Generated by nginx from `.env` at container start (section 8).
`/openmrs/spa/config/`	The frontend's `/config/` (logos, icons, translations).
`/openmrs/spa/*`	The team frontend container, full path passed through.
`/openmrs/*`	The team backend (OpenMRS), which reads the MySQL database and the Initializer content in `content/`.

The browser calls two services directly: the remote AI service (`https://emr-ai.20.84.70.78.nip.io`) and the local speech service (`http://localhost:8767`). The gateway's Content-Security-Policy allows both through `AI_CONNECT_SRC`.

3. Repository layout and ownership

`distro/` and `content/` are copies of other teams' repositories. Do not edit them here: fix things upstream, then refresh the copy.

Path	What it is	Owner
`distro/`	Team distro, unchanged. Refresh with a `git archive` of a newer main and update `PLATFORM_IMAGE_TAG`.	Distro team
`content/`	Initializer content snapshot (`ethiopiaemr-openmrs-content`). Changes go to that repository through a PR.	Content team
`docker-compose.platform.yml` , `gateway/` , `platform.ps1` , `scripts/` , `tools/build-spa-overlay.py`	The platform overlay.	This repo
`modules/<name>/module.json`	Where each extra module's build comes from.	This repo
`modules/esm-form-studio-app/`	Form Studio source (migrated 2026-10-22). Its `dist/` is git-ignored.	This repo
`modules/esm-ai-assistant-app/dist`	AI assistant built bundle 0.1.0. No source here; kept until Bini publishes the package.	Bini (AI service owner)
`services/asr-service/`	Local speech-to-text.	This repo
`tools/demo-video/`	Demo-video pipeline. Set `EMR_URL` to target another EMR.	This repo
`database/initial-db/`	Database dump, not versioned (`scripts/get-initial-db.ps1`).	Local only
`database/snapshots/` , `work/`	Snapshots and build output. Git-ignored.	Local only

4. Running it

Windows, PowerShell, from the repository root. `platform.ps1` is the one entry point; it wraps `docker compose --project-directory . -f distro/docker-compose.yml -f docker-compose.platform.yml`.

Quick start

1. `copy .env.example .env`, then set `AI_SERVICE_API_KEY` (ask the AI service owner). `platform.ps1` creates `.env` from the example if it is missing.
2. `.\platform.ps1 up` starts the stack. See the timings below.
3. `.\scripts\build-form-studio.ps1` makes the production build of Form Studio.
4. `.\platform.ps1 overlay` adds Form Studio and the AI assistant to the app. Refresh the browser.
5. Optional: `services\asr-service\start-asr.ps1` starts the speech service for Speak, in its own window.

Commands

Command	What it does
`up`	Starts the stack. On an empty install with a snapshot present it restores the snapshot first. Without a snapshot or a dump it runs `scripts/get-initial-db.ps1`.
`overlay`	Runs `tools/build-spa-overlay.py` and restarts the gateway. Run after start and after any module rebuild.
`status`	Containers and health (`docker compose ps`).

Command	What it does
`logs [service]`	Follows the logs of one service or all.
`stop`	Stops the containers and keeps data.
`down`	Removes the containers and keeps the data volumes.
`config`	Prints the merged compose file. Must succeed before a PR.
`content-pass`	Fresh install only, after the first start: clears the Initializer checksums and restarts the backend for a second content pass.
`snapshot`	Stops backend and db, saves the two data volumes to `database/snapshots/`, starts them again.

How long it takes

Measured on 2026-10-22.

Start	Time	Why
Normal restart (`stop` then `up`)	about 3.5 min	Data is kept in the Docker volumes.
Fresh install with a snapshot in `database/snapshots/`	**about 6 min**	`up` restores the loaded database and OpenMRS data.
Fresh install without a snapshot	about 1.5 h	Database import, search index over about 81,000 concepts, all content, then `content-pass`.

Snapshot procedure

1. Finish one full first start, including `.\platform.ps1 content-pass`.
2. Run `.\platform.ps1 snapshot`. It writes `mysql-data.tgz` and `openmrs-data.tgz` (about 900 MB together, git-ignored) and restarts backend and db (2 to 3 min).
3. Copy `database/snapshots/` to teammates. On their empty install, `up` finds both files and restores them before starting.

Content pass

The Initializer saves checksums even for content files that failed, and some files refer to concepts defined in later files. `content-pass` deletes `/openmrs/data/configuration_checksums` and restarts the backend so every file is processed again. Afterwards check:

```
docker logs ethioemr-platform-backend-1 2>&1 | Select-String 'was processed and [0-9]+ out of'
```

The expected result is no lines. Today about 19 files still appear (section 12).

5. Ports

The platform's ports are chosen not to clash with the other local stacks.

Port	Used by
`18200`	This platform: gateway, app at `/openmrs/spa` (`PLATFORM_HTTP_PORT`).
`3320`	This platform: MySQL (`PLATFORM_DB_PORT`).
`8767`	Speech service on `127.0.0.1` (`ASR_PORT`).
`18090` and `3309`	AI quick start (not this platform).

Port	Used by
`18100`	Form Studio's former workspace (not this platform).
`18080` and `3308`	The distro clone (not this platform).

6. Configuration (`.env`)

`.env` is git-ignored. `docker compose` reads it from the repository root. Add every new setting to `.env.example` with an empty or safe value.

Key	Value in `.env.example`	What it controls
`PLATFORM_IMAGE_NS`	`meronmen`	Docker namespace of the team images.
`PLATFORM_IMAGE_TAG`	`sha-9147129`	Image tag; matches the commit of `distro/`.
`PLATFORM_HTTP_PORT`	`18200`	Gateway port on the host.
`PLATFORM_DB_PORT`	`3320`	MySQL port on the host.
`AI_SERVICE_URL`	`https://emr-ai.20.84.70.78.nip.io`	AI service address given to the browser.
`AI_SERVICE_API_KEY`	(empty)	API key given to the browser. Ask the AI service owner.
`ASR_SERVICE_URL`	`http://localhost:8767`	Speech service address given to the browser.
`AI_CONNECT_SRC`	AI service, `http://localhost:8767`, `http://127.0.0.1:8767`	Extra addresses allowed in the CSP `connect-src`.
`SHR_PULL_PASSWORD`	(empty)	Optional SHR pull account password (team compose). Leave empty locally.

What `docker-compose.platform.yml` changes on top of the team compose file, and why:

Service	Change
project	Own compose project `ethioemr-platform`: containers, volumes and network never touch another stack.
gateway	Built from `distro/gateway`; `platform.conf.template` and `work/spa-overlay` mounted; AI and speech variables passed in; port `18200:80`.
frontend, backend	Team images pinned to the `distro/` commit, not `latest`. The backend reads `./content/configuration/backend_configuration`.
db	MySQL 8.0.45, because the v8 dump does not load on 8.0.46. Initialised from `database/initial-db/`.

7. Extra frontend modules (the overlay)

The published frontend image does not contain Form Studio or the AI assistant. Until each is published to npm and listed in the distro, `tools/build-spa-overlay.py` adds them:

1. It reads the frontend container's own `importmap.json` and `routes.registry.json`.
2. For each `modules/<name>/module.json` it copies the built files to `work/spa-overlay/modules/<name>/` (source maps skipped unless `--keep-maps`).
3. It adds one import map entry per module, pointing at `./overlay/<name>/<entry>`, and adds the module's `routes.json` to the route registry. App shell 10.x keeps routes under a top-level `routes` object; the script handles both layouts.

4. The gateway serves the merged files in place of the frontend's, with `Cache-Control: no-store`, so a rebuild shows on refresh.

`module.json` key	Meaning
`package`	npm package name, for example `@palladium-ethiopia/esm-form-studio-app`.
`dist`	Folder with the build: relative path, absolute path, or `$ENV_VAR`.
`entry`	Entry file. Optional; defaults to `<scope>-<name>.js`.

Adding a module

1. Build it (production). The build must contain `routes.json`, or the script stops with "no routes.json".
2. Add `modules/<name>/module.json` with `package`, `dist` and `entry`. Copy one of the two existing files.
3. Run `.\platform.ps1 overlay` and refresh the browser. Check that the browser console has no new errors.

The team's long-term route is to publish the module to npm and add one line to `distro/frontend/spa-assemble-config.json` in the distro repository. After that the overlay is no longer needed for it.

8. AI assistant configuration in the gateway

The AI assistant needs three values in the browser: the AI service address, its API key and the speech service address. They come from `.env`, so the key is not stored in any repository or image.

The config file

nginx answers `/openmrs/spa/ai-assistant.config.json` itself. The template fills it from the gateway's environment when the container starts:

```
{"@palladium-ethiopia/esm-ai-assistant-app":

{"aiServiceUrl":"${AI_SERVICE_URL}","aiServiceApiKey":"${AI_SERVICE_API_KEY}","asrServiceUrl":"${ASR_SERVICE_URL}"}}
```

Why a `sub_filter` is needed

The app shell loads the config files listed in `index.html`. The published frontend image fixes that list at build time, so the `SPA_CONFIG_URLS` variable in the compose file is ignored. The gateway therefore edits the page as it passes through:

```
proxy_set_header Accept-Encoding "";
sub_filter_once on;
sub_filter 'configUrls: [' 'configUrls: ["/openmrs/spa/ai-assistant.config.json", ';
```

Detail	Reason
Placed first in the list	Later files win. The AI file sets only the assistant's own keys, so the team's files keep the last word.
`Accept-Encoding ""`	`sub_filter` can only edit an uncompressed response, so the gateway asks the frontend for one.
CSP `connect-src`	The browser may call only the addresses in `AI_CONNECT_SRC` besides the EMR itself.

Check it: `http://localhost:18200/openmrs/spa/ai-assistant.config.json` returns the three values (the key included, see section 13).

9. Speech service

The **Speak** button needs speech-to-text, and the EMR images do not include it. `services/asr-service` is that service, run locally on `127.0.0.1:8767`. Audio never leaves the machine.

Endpoint	Behaviour
`GET /health`	`{"status":"ok","model":"..."}`. The assistant then shows "Speech service ok" with the model name.
`POST /transcribe`	Form with `audio` (WAV) and `lang` (`en` or `am`). Returns `text`, `model`, `processing_time`; for Amharic also what each engine heard and `english_words_used`.

Start it

`.\start-asr.ps1` in `services/asr-service`. The first run creates `.venv` and installs about 3 GB of Python packages; the models (about 3.5 GB) download into the shared Hugging Face cache on first use. Leave the window open. After the platform's first start, rebuild the medical vocabulary from the EMR database with `python build-lexicon.py`.

Mixed Amharic and English

In `mixed` mode (the default) two engines run on the same recording and `mix.py` merges them word by word, without an LLM. The Amharic engine hears everything but writes English words in Amharic letters. An English word from Whisper replaces what the Amharic engine heard at the same moment only if Whisper is reasonably sure, it sounds like the Amharic engine's word, and it is in the EMR's own vocabulary (`lexicon\emr-vocab.txt`, about 29,000 words from concept and drug names). Every output word comes from one of the two engines.

Setting	Default in `server.py` and `start-asr.ps1`	Meaning
`ASR_MODEL`	`mobiuslabsgmbh/faster-whisper-large-v3-turbo`	English Whisper model (lead).
`ASR_CONFIRM_MODEL`	`small`	Second English model that confirms words; empty turns it off.
`ASR_AM_MODE`	`mixed`	`mixed` or `transcribe` (Amharic engine only).
`ASR_DEVICE`	`auto`	`auto`, `cuda` or `cpu`.
`ASR_SAVE_DIR`	(off)	Saves each recording; `-SaveRecordings` sets it.
`ASR_PORT`	`8767`	Port.

The Amharic engine is `agkphysics/wav2vec2-large-xlsr-53-amharic`: 9% character errors on the 15 s test consultation, 3 s on CPU. Whisper large-v3 alone gave 90% character errors on Amharic and was rejected.

Amharic needs a setting. The assistant's `asrLanguage` defaults to `en`. Set it to `am` in Implementer Tools; it is stored in the browser only (local storage key `openmrs:temporaryConfig`).

10. Form Studio build

`scripts/build-form-studio.ps1` makes the production build of `modules/esm-form-studio-app` into its `dist/`, where the overlay reads it.

Command	What it does
`.\scripts\build-form-studio.ps1`	Installs packages if `node_modules` is missing (`--immutable`), then builds.
`.\scripts\build-form-studio.ps1 -Check`	Runs unit tests, type check and lint first; stops at the first failure.
First time, by hand	`cd modules\esm-form-studio-app; node .yarn/releases/yarn-4.10.3.cjs install`

The script writes `dist\BUILD-INFO.txt` with the build time and commit, then prints "Done. Next: .\platform.ps1 overlay". Yarn 4.10.3 is committed in `.yarn/releases`, so no global Yarn is needed.

11. Testing

Results below were run on 22 October 2026 against this platform, unless marked otherwise.

Check	Command (in `modules\esm-form-studio-app`)	Result
Unit tests	`node .yarn/releases/yarn-4.10.3.cjs test`	100 passed, 6 skipped (106) in 32 files; 143 s.
Type check	`node .yarn/releases/yarn-4.10.3.cjs typescript`	No errors; 32 s.
Lint	`node .yarn/releases/yarn-4.10.3.cjs lint`	No errors; 94 s.
End-to-end UI	`node node_modules\@playwright\test\cli.js test`	43 passed in 9.8 min: 42 tests and 1 setup step in 11 files, against `:18200`.
Compose file	`.\platform.ps1 config` (repo root)	Must succeed before a PR.
Speech merge	`python test-merge.py` (in `services/asr-service`)	Must still show "(no swaps)" on `test-audio/consult-am.wav`.

End-to-end suite

Playwright drives the installed Chrome through Form Studio the way a form builder would. The harness refuses any address that is not localhost, runs one browser at a time, logs in once, closes the EMR's red "unresolved backend dependencies" pop-up and accepts "Leave site?" dialogs. The login lives in `e2e/.env` (copy `e2e/example.env`; git-ignored). Reports go to `work/e2e-report/`.

The test plan's gates G0 to G3 passed (last on 2026-10-20). Specs 05 (editor) and 06 (rules) exist and passed in the run above, once; gate G4 asks for three runs in a row and has no recorded result yet. Spec `08-check-export` (the Check and the download) is not written.

12. Known gaps

Gap	Detail	Owner
Content rows left unsaved	After the content pass about 19 content files still leave rows unsaved, for example `queue.csv` 18 of 18 and `radiology_category_members_concept_set.csv` 101 of 232. A separate install on the same images and content shows the identical list, so they come from the content snapshot.	Content team

Gap	Detail	Owner
AI bundle built for app shell 9.x	The bundle in `modules/esm-ai-assistant-app/dist` (0.1.0) was compiled for 9.x; the team runs 10.0.1-pre.5263. On 22 Oct all four features worked on 10.0.1 (see below). It still needs a 10.x build, an npm package and a shared source repository.	Bini
AI service certificate	The AI service's TLS certificate is not trusted by browsers. Until a user has accepted the warning once, Form Filler falls back to its rule-based extractor ("AI service not reachable"), observed on 22 Oct as `net::ERR_CERT_AUTHORITY_INVALID`.	Bini
Upload input	Form Filler and Diagnosis Assistant show **Upload** marked "SOON". It is not available.	Bini
Orders not sent to Diagnosis	The Diagnosis Assistant says "Not loaded: orders. The model will not see them."	Bini
Form Studio back end	There is no back end yet; Form Studio runs in the browser only.	This repo

AI assistant live check, 22 October 2026

Done on `:18200` with the synthetic patient Almaz Bekele and the **Central Triage Form**, driven by Playwright in Chrome.

Feature	Request	Result
Help & Search	`POST /v1/help/answer?stream=true` : 200	Answer with 5 numbered sources (4 forms, 1 WHO guideline page); model `qwen2.5:14b`.
Form Filler	`POST /v1/form/extract` : 200	8 values found from a typed note, each with its quoted evidence; **Insert accepted (8)** filled the form fields. Not saved.
Diagnosis Assistant	`POST /v1/diagnosis?stream=true` : 200	Three-item differential from 8 chart records, with guideline references and a "Some citations could not be verified" warning.
Speak	`POST localhost:8767/transcribe` : 200	English (`en`) and Amharic (`am`) recordings transcribed from a test WAV file played as the microphone.

13. Security notes

> CAUTION
> **Any user who can open the EMR can read the AI service API key in the browser.** The gateway keeps it out of git and images, but `/openmrs/spa/ai-assistant.config.json` returns it to every page, like any frontend configuration.

Open items for the AI service owner, most urgent first:

#	Item	Question
1	Rotate the key	It was distributed in the AI quick-start zip and images. What can a holder of the key do: which endpoints, what rate limit, can it reach endpoints that take patient context?
2	Canonical address and TLS	Config uses `emr-ai.20.84.70.78.nip.io`, the integration docs say `ai.20.84.70.78.nip.io`. Which is canonical, and can it get a trusted certificate?
3	`CORS_ORIGINS`	Which origins are allowed? Calls from `http://localhost:18200` already succeed, which suggests a wildcard or no origin check.
4	User identity	`X-OpenMRS-User` is trusted but not verified. When does the same-origin route `/openmrs/ai/` with OpenMRS session validation arrive? It removes the key from browsers.

#	Item	Question
5	Patient data handling	Are prompts, chart records or answers logged? Where, for how long, and who can read them?
6	Frontend on shell 10.x	Rebuild against 10.0.1-pre.5263 with peer `10.x`, publish `@palladium-ethiopia/esm-ai-assistant-app` to npm, share the source repository.

Other rules: secrets (`.env`, keys) never go into git; the platform holds synthetic test data only; the speech service listens on `127.0.0.1` only and keeps audio on the machine; recordings saved with `-SaveRecordings` must be test dictations only.

14. Conventions

From `CONTRIBUTING.md`, following the team distro.

Topic	Rule
Branches	`main` is the integration branch; never commit to it. Branch from it with `feat/`, `fix/`, `chore/` or `docs/`. One topic per branch and per PR.
Merges	PRs are merged with a merge commit ("Merge pull request #N from ...").
Commits	Conventional Commits: `feat: ...`, `fix(gateway): ...`, `chore: pin distro to sha-9147129`. Imperative, present tense, about what changes and why.
Other teams' code	`distro/` and `content/` are copies. Fix upstream, then refresh.
Before a PR	`.\platform.ps1 config` succeeds; after overlay or module changes, the browser console is free of new errors; after speech changes, `test-merge.py` shows "(no swaps)"; the PR says where a reviewer can see anything new.

15. Where to see the work

What	Code	Where to look	How to verify
Platform stack	`docker-compose.platform.yml`, `gateway/`, `platform.ps1`	`http://localhost:18200/openmrs/spa`	`.\platform.ps1 status` shows 4 containers (gateway, frontend, backend, db).
Overlay	`tools/build-spa-overlay.py`, `modules/*/module.json`	`/openmrs/spa/importmap.json`	Lists both `@palladium-ethiopia` modules with paths under `./overlay/`.
Form Studio	`modules/esm-form-studio-app`	`/openmrs/spa/form-studio`	The start page lists the forms; `build-form-studio.ps1 -Check` passes.
AI assistant	`modules/esm-ai-assistant-app/dist`	Any patient chart, Alt+A	Help & Search answers with sources.
Speech service	`services/asr-service`	`http://localhost:8767/health`	Returns `"status":"ok"`.
These manuals	`docs/platform/src/`	`docs/platform/*.pdf`	`node docs/platform/src/build-pdfs.mjs` rebuilds both.